

Adolescent BMI Dynamics: Age, Sex, & Growth Trajectories

工管二乙 113370202 方立謙

Foundation & Main Model

Data: YRBS 2007

1. Research Question

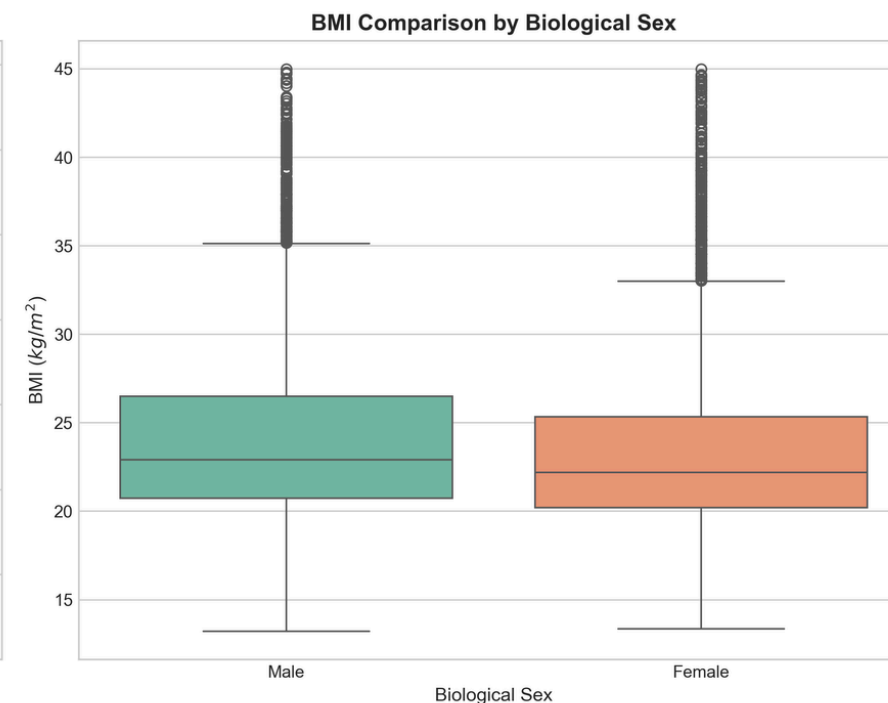
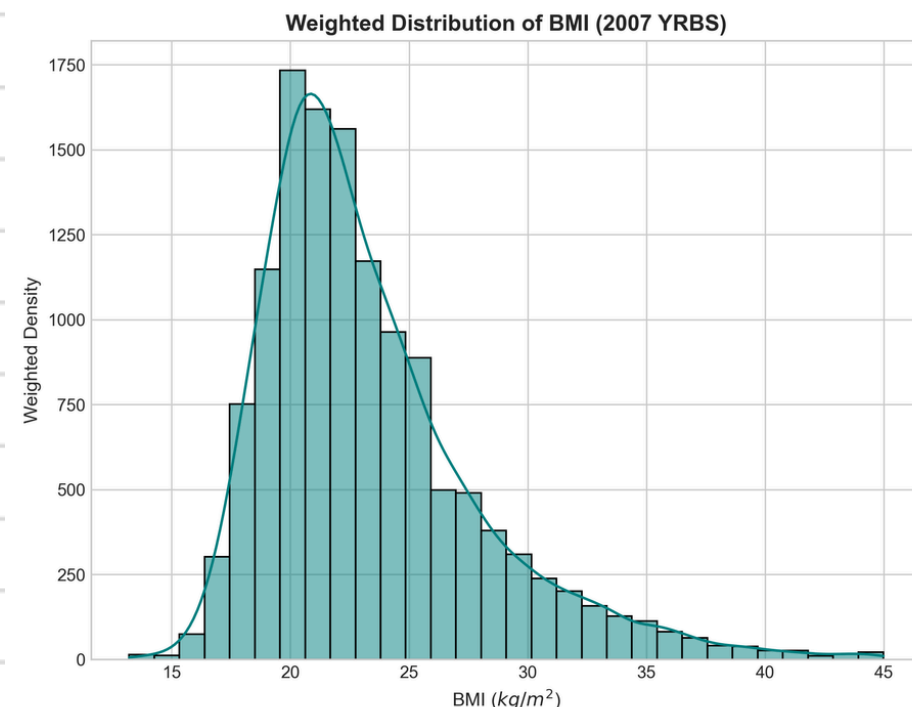
Do Age and Sex predict adolescent BMI, and do growth rates differ by gender?

2. Variables & Data Cleaning

- Dependent (Y): BMI (kg/m^2) (Derived from Height & Weight)
- Independent (X): Age (12-18) & Sex (Female = Ref / Male)
- Methodology: Weighted Least Squares (WLS) (Using survey weights)
- Data Cleaning: Excluded missing data & BIV (BMI < 12 or > 45)
Final N = [number of sample]

3. Main Effects Model

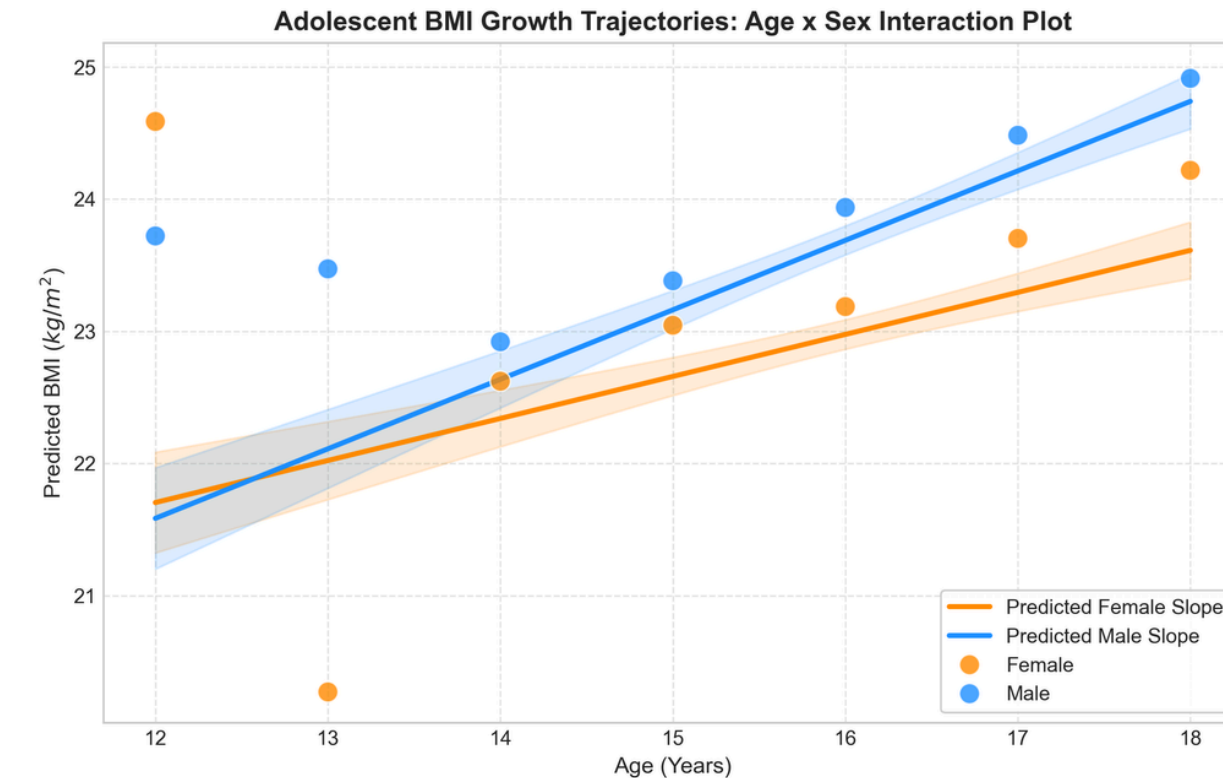
- Equation: $\text{BMI} = 13.82 + 0.45 \cdot (\text{Age}) + 0.32 \cdot (\text{Is_Male})$
- Findings ($p < 0.001$):
- Age: Each year adds $+0.45 \text{ kg/m}^2$ to BMI.
- Sex: Males average $+0.32 \text{ kg/m}^2$ higher than females.



Interaction & Summary

4. Deep Dive: Interaction Effect

- Equation: $\text{BMI} = 14.12 + 0.35 \cdot (\text{Age}) - 1.12 \cdot (\text{Male}) + 0.13 \cdot (\text{Age} \times \text{Male})$
- Slope Comparison ($p < 0.05$):
- Female Growth: $+0.35 \text{ kg/m}^2 / \text{year}$
- Male Growth: $+0.48 \text{ kg/m}^2 / \text{year}$ (Significant acceleration)
- Key Takeaway: Males experience a significantly faster velocity of BMI growth during puberty.



Diagnostics & Conclusion

- Diagnostics: Homoscedasticity and normality of residuals were verified; large sample (N) ensures robust asymptotic inference.
- Executive Summary: Adolescent BMI growth is highly dynamic. Biological sex and age interact: male high schoolers show significantly steeper BMI acceleration over time compared to female peers.